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Transmission and method for control of boost spool

Abstract

A gas turbine engine includes a first spool associated with a primary combustor, a second spool associated with a secondary combustor, and a third spool, each spool including a compressor and a turbine mounted to a shaft. A transmission and accessory gearing are enclosed within a housing of an accessory gearbox. The transmission rotationally couples the third spool to the second spool and accessory gearing. A method of operating the gas turbine engine includes supplying a first fuel flow rate to the primary combustor and supplying a second fuel flow rate to the secondary combustor within an intermediate speed range of the gas turbine engine.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) (1) This is a divisional of U.S. patent application Ser. No. 17/662,204, entitled “TRANSMISSION AND METHOD FOR CONTROL OF BOOST SPOOL,” filed May 5, 2022.

BACKGROUND

(1) The present disclosure relates generally to gas turbine engines and, more particularly, to gas turbine engines capable of operating in a high overall pressure ratio (OPR) mode and in a low OPR mode to adapt to the ambient conditions and to provide more efficient operation without exceeding thermal limits of the gas turbine engine.

(2) The overall pressure ratio (OPR) is a measure of the total pressure rise in a gas turbine engine (i.e., a pressure ratio equal to the air pressure discharged from the last compressor stage and the ambient air pressure entering the engine). Generally speaking, as OPR increases, the thermodynamic efficiency of the gas turbine engine increases, enabling the engine to consume less fuel per unit of thrust (i.e., thrust specific fuel consumption or TSFC) than a corresponding engine with lower OPR. However, air temperatures within the gas turbine engine increase with increasing

OPR and can produce temperatures within the compressor section and/or turbine section that exceed permissible material and structural limits. Furthermore, the maximum temperature within the compressor and the turbine increase as the ambient temperature increases, adding to the temperature increase associated with the OPR of the engine.

(3) Conventionally, turbine temperatures are maintained within acceptable limits by limiting OPR to a ratio that produces acceptable turbine temperatures for worst case ambient conditions, typically, design conditions corresponding to hot day take-off. While this technique produces a gas turbine engine design that provides an acceptable compromise for a variety of operating conditions, limiting OPR for hot day take-off conditions produces a gas turbine engine that operates at less OPR than otherwise possible at cruise power, reducing engine efficiency when high efficiency, low fuel consumption operation is most advantageous to extend aircraft range or payload capacity.

SUMMARY

(4) A gas turbine engine in accordance with an exemplary embodiment of this disclosure includes spool, a second spool, a third spool, an accessory gearbox, and a plurality of engine accessories. The first spool includes a first compressor and a first turbine mounted to a first shaft. The second spool includes a second compressor and a second turbine mounted to a second shaft. The third spool includes a third compressor and a third turbine mounted to a third shaft. The accessory gearbox encloses a transmission gearing and accessory gearing. Tower shaft rotationally couples the third spool and the transmission gearing. The transmission gearing rotationally coupling the third spool to the second spool and the accessory gearing.

(5) A further embodiment of the gas turbine engine in accordance with this disclosure includes a primary combustor and a secondary combustor. The primary combustor is disposed between and communicates with the first compressor and the first turbine. The secondary combustor is disposed between and communicates with the second compressor and the second turbine.

(6) A method of operating the foregoing gas turbine includes supplying a first fuel flow rate to the primary combustor contemporaneously with supplying a second fuel flow rate to the secondary combustor based on a power lever angle indicative of an intermediate power range of the gas turbine engine. The intermediate power range of the gas turbine engine is between a maximum continuous power level and a minimum continuous power level associated with an operational power range of the gas turbine engine.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a schematic representation of a gas turbine engine that includes a boost spool.

(2) FIG. 2 is a schematic depicting inlet and outlet duct assemblies for the boost spool

(3) FIG. 3 is a schematic depicting diffuser and primary combustor of the gas turbine engine.

(4) FIG. 4 is schematic of diffuser inlet and outlet areas of the inlet and outlet duct assemblies.

(5) FIG. 5 is a T-s diagram illustrating the thermodynamic performance of a gas turbine engine operating with a boost spool relative to a gas turbine engine operating without a boost spool.

(6) FIG. 6 is a schematic of a controller for the gas turbine engine.

(7) FIG. 7A and FIG. 7B are schematic of a transmission with multiple clutches for driving boost spool and accessory gearbox.

(8) FIG. 8 is an exemplary speed profile for a gas turbine engine with a transmission in accordance with FIG. 7A and FIG. 7B.

DETAILED DESCRIPTION

(9) As described herein, a gas turbine engine has a boost spool that can be selectively engaged to increase overall pressure ratio (OPR) during certain engine power levels (e.g., cruise power) while operating the gas turbine engine without the boost spool during other power levels (e.g., takeoff

power). A transmission rotationally couples boost spool to a low pressure spool of the engine and/or to the accessory gearbox facilitating improved speed profiles for the accessory gearbox. With this arrangement, the gas turbine engine can operate within thermal limits when ambient conditions limit the OPR and can operate with greater engine efficiency when ambient temperatures are lower and permit higher OPR operation.

(10) FIG. 1 is a schematic representation of gas turbine engine **10** that includes boost spool **12** in accordance with an exemplary embodiment of this disclosure. Gas turbine engine **10** is a dual spool engine that includes low pressure spool **14** and high pressure spool **16**. Low pressure spool **14** includes low pressure compressor **18** mechanically and rotationally connected to low pressure turbine **20** by shaft **22**, and high pressure spool **16** includes high pressure compressor **24** mechanically and rotationally connected to high pressure turbine **26** by shaft **28**. Bearings **30** and **32** support shaft **22** of low pressure spool **14**, and bearings **34** and **36** support shaft **28** of high pressure spool **16**, each at forward and aft shaft ends, respectively. Low pressure spool **14** and high pressure spool **16** are coaxial, each extending along and rotating about centerline **37** independently of one another.

(11) Compressors and turbines **18**, **20**, **24**, and **26** include at least one compressor stage or turbine stage, each stage formed by a row of stationary vanes and a row of rotating blades. In the exemplary embodiment depicted by FIG. 1, each of low pressure compressor **18** and high pressure compressor **24** has three stages, and each of low pressure turbine **20** and high pressure turbine **26** has two stages, although the number of stages in each compressor or turbine can be selected based on the desired pressure ratios as is known in the art.

(12) At times, boost spool **12**, low pressure spool **14**, and high pressure spool **16** may be referred to as a first spool, a second spool, and/or a third spool in which “first”, “second”, and “third” correspond to one of boost spool **12**, low pressure spool **14**, and high pressure spool **16**. Similarly, “first”, “second”, and/or “third” labels may be used in conjunction with corresponding components of the first spool, the second spool, and/or the third spool in order to distinguish components of each spool from components of the other spools.

(13) As shown in FIG. 1, at least one vane stage of low pressure turbine **20** includes variable area turbine (VAT) **38**. Variable area turbine **38** includes a row of vanes, each vane rotatable about a vane axis extending in a spanwise direction of the vane. The open area through variable area turbine (VAT) **38** changes depending on the stagger angle of vanes with respect to centerline **37**. The closed position occurs when vanes form a maximum stagger angle with respect to centerline **37** while the open position occurs when vanes form a minimum, and sometimes negative, stagger angle with respect to centerline **37**. The minimum open area typically coincides with the closed position since vanes tend to rotate toward each other, and in some instances vanes overlap when viewed along centerline **37**. As vanes move from the closed position towards the open position, the open area through the vane stage increases until a maximum open area is reached, typically near a minimum turning angle, or zero stagger angle position. In some embodiments, the open position coincides with the vane position associated with a maximum open area through the vane row. In other embodiments, vanes can continue to rotate towards the open position in which the vane stagger angle is negative, tending to decrease the open area as the stagger angle becomes more negative. A neutral position or nominal position of vanes can be associated with an angular vane position between the open position and the closed position that achieve a desired incident angle with a rotor of low pressure turbine **20**.

(14) Gas turbine engine **10** also includes fan **39** mounted to fan shaft **40**. One or more bearings **42** support fan shaft **40**, which is mechanically and rotationally coupled to low pressure spool **14**. Fan shaft **40** may be directly connected to shaft **22** of low pressure spool **14**. With this arrangement, fan **39** and fan shaft **40** rotate at the same speed and in the same direction as low pressure spool **14**. In other embodiments, such as the exemplary embodiment depicted in FIG. 1, fan shaft **40** may be rotationally coupled to shaft **22** via gearing **44**. For instance, gearing **44** can be an epicyclic gear

train that includes a central sun gear mounted to shaft **22**, a ring gear mounted to fan shaft **40**, and a plurality of planet gears circumferentially spaced about the sun gear and mechanically engaging the ring gear and the sun gear, the planet gears being supported by a planet carrier (not shown). Generally, gas turbine engines utilizing epicyclic gearing to drive fan **39** and fan shaft **40** restrain the planet carrier to cause fan shaft **40** to rotate slower (and in the opposite direction) than low pressure spool **14**. Accordingly, fan **39** and low pressure spool **14** can rotate at speeds that are more efficient for respective blade geometries.

(15) In operation, nose cone **46** guides ambient air flow **48** into inlet **50**. Rotation of fan **39**, which includes circumferentially spaced fan blades **52**, compresses ambient air flow **48** before splitter **54** divides flow **48** into bypass flow **56** and core flow **58**. Bypass flow **56** passes through bypass duct **60** to structural guide vanes **62** and discharges from engine **10** through a bypass flow exhaust nozzle (not shown), which is downstream from structural guide vane outlet **64**. Inlet guide vanes **66** guide core flow **58** into low pressure compressor **18** that subsequently flows into high pressure compressor **24**, each compressor stage further compressing core flow **58**. Compressed core flow **58** discharges from high pressure compressor **24** into diffuser **68**. Diffuser **68** fluidly connects high pressure compressor **24** to combustor **70** and includes divergent walls that reduce core flow **58** velocity and thereby increase static pressure of flow **58** before entering combustor **70**. Combustor **70** can be an annular combustor (or another suitable design). Fuel injected into combustor **70** mixes with compressed core flow **58**, and one or more ignitors combust the fuel-to-air mixture to produce a compressed and heated core flow **58** that is discharged into high pressure turbine **26**. Core flow **58** interacts with vanes and blades of high pressure turbine **26** causing rotation of shaft **28** about centerline **37** and driving rotation of high pressure compressor **24**. Similarly, core flow **58** interacting with vanes and blades of low pressure turbine **20** cause rotation of shaft **22** about centerline **37** to drive rotation of low pressure compressor **18** as well as fan shaft **40** directly or via gearing **44**. Downstream of low pressure turbine **20**, core flow **58** discharges from engine **10** through exhaust nozzle **72**.

(16) Boost spool **12** includes at least boost compressor **74** and shaft **76** fluidly connected to gas turbine engine **10** by inlet duct assembly **78** and outlet duct assembly **80**. In some embodiments, boost spool **12** also includes one or more of boost turbine **82**, combustor **84**, and variable inlet guide vanes **86**. Boost compressor **74** and boost turbine **82** include at least one compressor stage or turbine stage, each stage formed by a row of stationary vanes and a row of rotating blades. Variable inlet vanes **86** form an array of circumferentially spaced vanes at an inlet to boost spool **12** and upstream of boost compressor **74**. Each vane of variable inlet guide vanes **86** is rotatable about a vane axis that extends in a spanwise direction of the vane. An angular position of variable inlet guide vanes **86** ranges between a closed position, a neutral or nominal position, and an open position in the same manner as vanes of variable area turbine **38**. During operation of boost spool **12**, variable inlet vanes **86** can pivot to decrease or increase the open inlet area in order to vary the amount of core flow **58** diverted into boost compressor **74** through inlet duct assembly **78**. Shaft **76** mechanically and rotationally connects boost compressor **74** to boost turbine **82**, each component arranged coaxially with boost axis **88**. Bearings **90** and **92** support boost spool **12** with respect to a stationary casing, which may be affixed or incorporated to a casing of gas turbine engine **10** (e.g., casing **93** as depicted in FIGS. 7A and 7B). Transmission **400** mechanically and rotationally couples boost spool **12** to one or more spools of gas turbine engine **10** (e.g., low pressure spool **14** and/or high pressure spool **16**) as discussed further below.

(17) The position and orientation of boost spool **12** relative to gas turbine **10** is selected base on the particular details of the mechanical coupling to gas turbine engine **10**. Boost axis **88** can be parallel and offset from centerline **37** of gas turbine engine **10** as schematically shown by FIG. 1. Furthermore, FIG. 1 shows boost spool **12** with a reverse flow orientation (i.e., aft-to-forward flow) such that a flow direction through boost spool **12** from compressor **74** to turbine **82** is opposite a flow direction (i.e., forward-to-aft flow) through gas turbine engine **10** from inlet **50** to outlet **64**

and from inlet **50** to nozzle **72**. Alternatively, boost axis **88** can be oblique or perpendicular to centerline **37**.

(18) For all mounting positions of boost spool **12**, the location and orientation of boost spool **12** permits boost spool **12** to receive a compressed air flow from gas turbine engine **10** and to discharge an expanded air flow to gas turbine engine **10**. Boost spool **12** can receive a compressed airflow from any compressor stage of gas turbine engine **10** to achieve varying degrees of boost compression. In one exemplary embodiment, boost spool **12** receives a compressed air flow from a location that is downstream from the last compressor stage of the gas turbine engine. In the case of gas turbine engine **10**, boost spool **12** receives airflow from diffuser **68** and discharges an expanded airflow to diffuser **68**. In other instance, boost spool receives airflow from diffuser **68** and discharges an expanded airfoil to both diffuser **68** and combustor **70**, which is downstream of high pressure compressor **24** and upstream from high pressure turbine **26**.

(19) In operation, boost spool **12** receives a portion of core flow **58** extracted from diffuser **68** (i.e., boost flow **94**) and routed to an inlet of boost compressor **74** through inlet duct assembly **78**. Within boost compressor **74**, the pressure and temperature of boost flow increases with each compressor stage. Compressed boost flow **94** enters combustor **84** where injected fuel mixes with compressed boost flow **94**. Once the fuel-air mixture is ignited, boost flow **94** discharges into boost turbine **82**. Turbine **82** expands boost flow **94** across each turbine stage, driving turbine **82**, shaft **76**, and compressor **74**. Expanded boost flow **94** discharges from boost spool **12** through outlet duct assembly **80**, which may route discharged air to diffuser **68**, combustor **70**, or both diffuser **68** and combustor **70**.

(20) FIG. 2 schematically depicts inlet duct assembly **78** that extracts a portion of core flow **58** from diffuser **68** and outlet duct assembly **80** that discharges the boost flow to diffuser **68** and combustor **70**. As shown, diffuser **68** includes inner peripheral wall **96** and outer peripheral wall **98** spaced radially outward from wall **96**. Multiple struts **100** extend from inner peripheral wall **96** to outer peripheral wall **98** of diffuser. FIG. 2 depicts five struts **100**. However more or less struts **100** can be used in other examples, each incorporating features of inlet duct assembly **78** and outlet duct assembly **80** discussed below. Combustor **70** includes outer casing **102** and spaced radially from inner casing **104** to define an annular combustion chamber. Inner casing **104** and outer casing **102** are thermally protected by segmented liners **106**.

(21) Depicted using solid lines, inlet duct assembly **78** includes multiple branch ducts **108** collected into inlet manifold **110**. Each branch duct **108** communicates with diffuser **68** via respective branch inlets **112**. In some embodiments, branch inlets **112** can be formed by inner and/or outer peripheral walls of diffuser **68** such that branch ducts **108** extract core flow **58** through inner and outer walls of diffuser **68**. In other embodiments, branch inlets **112** are formed by respective struts **100**. Branch inlet ducts **108** extend from branch inlets **112** to inlet manifold **110**. Inlet manifold **110** can be a pipe, duct, or plenum accommodating the collected flow through each branch inlet duct **108** and routing the accumulated inlet flow to inlet of boost spool **12**.

(22) Depicted using dashed lines, outlet duct assembly **80** can include one or more ducts extending from an outlet of boost turbine **82** to diffuser **68**, combustor **70**, or both diffuser **68** and combustor **70**. In one exemplary embodiment, outlet duct assembly **80** can include main duct **114** extending from an outlet of boost turbine **82** to one or more branch outlet ducts **116**, one or more branch outlet ducts **116**, or one or more branch outlet ducts **116** and one or more branch outlet ducts **118**. Each branch outlet duct **116** extends from main duct **114** to one of boost outlets **120** formed in an inner peripheral wall or an outer peripheral wall of diffuser **68**, or a wall of strut **104**, placing main duct **114** and boost turbine **82** in communication with diffuser **68**. Each branch outlet duct **118** extends from main duct **114** to one of boost outlets **122** formed in a peripheral wall of combustor **70**, placing main duct **114** and boost turbine **82** in communication with combustor **70**.

(23) As shown in FIG. 2, outlet duct assembly **80** includes main duct **114**, multiple branch ducts **116**, and branch duct **118**. Main duct **114** extends from boost turbine **82** to each branch outlet duct

116 and branch duct **118**. Branch outlet ducts **116** extend from main duct **114** to boost outlets **120** formed in walls of respective struts **100**. Branch outlet duct **118** extends from main duct **114** to boost outlet **122** formed in a peripheral wall bounding a combustion zone of combustor **70**.

(24) FIG. **3** is a schematic of diffuser **68** and combustor **70** of gas turbine engine **10**. As shown, branch inlet duct **108** extracts air from core flow **58** through inlet **112**, and branch duct **116** discharges boost exhaust flow through outlet **120**. Branch inlets **112** and branch outlets **120** are formed by strut **100**. One or both of branch inlet **112** and branch outlet **120** can extend from inner peripheral wall **96** to outer peripheral wall **98**. Outlet **122** of branch outlet duct **118** discharges through outer casing of combustor **70**. The location of outlet **122** is spaced axially downstream from injectors **124**, or between injectors **124** and high pressure turbine **26**.

(25) FIG. **4** depicts another schematic view of struts **100** and the flow distribution within diffuser **68**. Each strut **100** includes first side wall **126** and second side wall **128** extending from leading edge **130** of strut **100** to rear wall **132**. Inlets **112** are disposed in one of sidewalls **126** and **128** while outlets **120** are disposed in the other sidewall opposite the inlet sidewall. During operation of boost spool **12**, a portion of compressed core flow **58** enters each inlet **112** and flows through branch inlet ducts **108** and manifold **12** to the inlet of boost spool **12** and boost compressor **74**. A portion of boost spool flow exiting boost turbine **82** flows through each branch outlet duct **116** before discharging through each outlet **120** into diffuser **68**.

(26) An outlet flow division can be achieved with appropriate selection of length, cross-sectional area, and routing of main duct **114**, diffuser branch ducts **116**, combustor branch duct **118** and associated outlets **120** and **122** of outlet duct assembly **80**. A minimum mass flow rate of boost exhaust discharged to diffuser through outlets **120** relates to an amount of flow required to maintain flow into boost spool **12** through inlet duct assembly **78**. Adequate flow through inlet duct assembly **78** can be achieved by maintaining at least a minimum static pressure at branch inlets **112** throughout all operating conditions during which boost spool **12** can be operated, including transient periods associated with starting or stopping boost spool **12**. The maximum mass flow rate of boost exhaust discharged to diffuser **68** through outlets **120** relates to a maximum temperature of fuel injectors within combustor **70**. As boost exhaust mass flow rate increases, a temperature of compressed air entering combustor **70** increases. Accordingly, the maximum permitted temperatures of components of combustor **70** during continuous operation limits the maximum mass flow rate of boost exhaust returned to diffuser **68** through outlets **120**. Expressed as a percentage of total flow through boost spool **12**, five percent to forty percent of boost exhaust flow can be discharged into diffuser **68** while the remainder ninety-five percent to sixty percent of boost exhaust flow can be discharged to combustor **70**.

(27) The flow division of boost exhaust between diffuser **68** and combustor **70** allows boost outlets **120** to extend from the inner peripheral wall to the outer peripheral wall of diffuser **68** as shown in FIG. **3**. As the radial extent of boost outlets **120** approaches the full radial extent of diffuser **68** at struts **100**, flow uniformity and flow stability increase within diffuser **68** while component temperatures within combustor **70** remain acceptable for continuous operation.

(28) FIG. **5** is a T-s diagram illustrating the thermodynamic performance of gas turbine engine **10** boosted by spool **12** relative to gas turbine engine **10** operating without boost spool **12**. Entropy is displayed along abscissa axis **134**, and temperature is displayed along ordinate axis **136**, each increasing from origin **138**. Dashed curve A depicts the preferred thermodynamic cycle of gas turbine engine **10** operating without boost engaged at takeoff power. Dashed curve B depicts the thermodynamic cycle of gas turbine engine **10** were boost to be engaged while operating at takeoff power on a hot day. Solid curve C depicts gas turbine engine **10** operating without boost spool **12** engaged while operating at cruise power. Solid curve D depicts the preferred thermodynamic cycle of gas turbine engine **10** operating with boost engaged while operating at cruise power having an OPR (value=X) that is greater than the OPR of boosted gas turbine engine **10** were boost to be engaged while operating at takeoff power on a hot day.

(29) Each of curves A, B, C, and D are defined by points 2, 3, 4, and 5, respectively. Accordingly, dashed curve A extends from point 2A to point 3A, representing the compression work completed by engine 10 between engine inlet 50 and the exit of high pressure compressor 24 (see FIG. 1). From point 3A, dashed curve A extends along a line of constant pressure ratio (value=Y) to point 4A that represents the heat added to core flow 58 through combustion. After combustion, high pressure turbine 26 and low pressure turbine 20 extract work from the heated and compressed core flow 58, a process represented by dashed line A between points 4A and 5A. Dashed curve B extends between point 2B to point 3B during the compression phase, between point 3B and 4B along a line of constant pressure ratio (value=Z) during combustion, and between point 4B and 5B during turbine expansion. Solid curves C and D are defined by points 2C, 3C, 4C, and 5C and points 2D, 3D, 4D, and 5D in a similar manner to curves A and B. An engine operating along curve A has less OPR (value Y) than an engine operating on any of the other curves (i.e., curves C and D having an OPR equal to value X and curve B having an OPR value equal to Z). Moreover, an engine operating on curves C and D have an OPR value X that is greater than an engine operating on curve B with an OPR value Z.

(30) The temperature entering the compressor section of gas turbine engine 10 at cruise power is lower than the temperature entering the compressor section of gas turbine engine 10 at takeoff power because the ambient temperature at cruising altitude is lower than the ambient temperature during a hot day takeoff. For example, at cruising altitude, the ambient temperature can be approximately -26.1 degrees Celsius (or about -15 degrees Fahrenheit) while on a hot day takeoff, the ambient temperature can be approximately 46.1 degrees Celsius (or about 115 degrees Fahrenheit). For each curve, the temperature within the engine at points 3A, 3B, 3C, and 3D are limited to a line of constant temperature labeled "T3 Limit" while the temperature at points 4A, 4B, 4C, and 4D are limited to a line of constant temperature labeled "T4 Limit".

(31) Unboosted operation of gas turbine engine 10 represented by dashed curve A trades engine fuel efficiency and engine materials life between temperature limits during a hot day takeoff and cruising. As a result, the OPR of unboosted operation of engine 10 is reduced by hot day takeoff conditions (i.e., the temperature at point 4A is lower than the temperature at 4B). The area bounded by dashed curve A and a line connecting points 5A and 2A represent the amount of work completed by engine 10 while operating at takeoff power and without boost spool 12 operation. The area bounded by dashed curve B and a line connecting points 5B and 2B represent the amount of work completed by engine 10 while operating at takeoff power and with boost spool 12 operation. The two areas are the same and the amounts of work completed by the engine at takeoff are the same. The amount of heat energy rejected by gas turbine engine 10 while operating in accordance with dashed curve A is shown by horizontally-hatched area 140. The thermodynamic efficiency of gas turbine engine 10 operating in accordance with dashed curve A is the work energy divided by the summation of work and rejected heat energy defined by curve A.

(32) Contrastingly, the work performed by gas turbine engine 10 with the aid of boost spool 12 is bounded by curve D and a line extending between points 5D and 2D while the heat energy rejected by boosted gas turbine engine 10 operating at cruise power is shown by vertically-hatched area 142. Regions where areas 140 and 142 overlap appear as a square-hatched area. By comparing the sizes of work areas bounded by curves D and C relative to heat rejection areas 140 and 142, respectively, it is evident that work area D represents a larger percentage of the total area under curve D than corresponding areas under curve C. Accordingly, operating gas turbine engine 10 with boost spool 12 at cruise power results in more efficient thermodynamic operation and, thus, improved thrust specific fuel consumption (TSFC) than operating engine 10 without boost spool 12. Furthermore, a gas turbine engine with the same OPR as boost engine operation depicted by curve D does not have the improved engine fuel efficiency and same life of operation. As shown in FIG. 5 (i.e., the temperature at 4B is higher than the temperature 4A). As such, gas turbine engine 10 can be operated without boost spool 12 during hot day takeoff conditions (i.e., dashed curve A)

and can be operated with boost spool **12** at cruise power (i.e., solid curve D) to achieve greater thermal efficiency at cruise power while satisfying thermal limits for hot day takeoff conditions. (33) FIG. **6** is a schematic of controller **150** that regulates the operation of gas turbine engine **10** and, more particularly, coupling and decoupling of boost spool **12** via one of transmission **400**. Additionally, controller **150** regulates fuel flow rates to primary combustor **70** and secondary combustor **84** based on one or more engine parameters, aircraft parameters, and/or exterior conditions. Controller **150** can include a standalone control unit or a control module incorporated into another control unit. Furthermore, controller **150** can be an amalgamation of distinct control units and/or distinct control modules that together perform the functions described in this disclosure. In some embodiments, controller **150** can be a full authority digital engine control (FADEC), an electric engine controller (EEC), or an engine control unit (ECU).

(34) Controller **150** includes processor **152**, memory **154**, and input/output interface **156**. Processor **152** executes one or more control algorithms **158** stored within memory **154** to output engine control signals **160** based on one or more input signals **162**. Examples of processor **152** can include any one or more of a microprocessor, a controller, a digital signal processor (DSP), an application specific integrated circuit (ASIC), a field-programmable gate array (FPGA), or other equivalent discrete or integrated logic circuitry.

(35) Memory **154** can be configured to store information within controller **154**. Memory **154**, in some examples, is described as computer-readable storage media. In some examples, a computer-readable storage medium can include a non-transitory medium. The term “non-transitory” can indicate that the storage medium is not embodied in a carrier wave or a propagated signal. In certain examples, a non-transitory storage medium can store data that can, over time, change (e.g., in RAM or cache). Memory **154** can include volatile and non-volatile computer-readable memories. Examples of volatile memories can include random access memories (RAM), dynamic random-access memories (DRAM), static random-access memories (SRAM), and other forms of volatile memories. Examples of non-volatile memories can include, e.g., magnetic hard discs, optical discs, flash memories, or forms of electrically programmable memories (EPROM) or electrically erasable and programmable (EEPROM) memories.

(36) Input/output interface or I/O interface **156** can be a series of input and output channels that electrically communicate with an engine control bus. The engine control bus interconnects controller **150** with various components of the gas turbine engine **10** described above such that engine control signals **160** can be transmitted to individual engine components and input signals **162** can be received.

(37) Engine control signals **160**, input signals **162**, or both engine control signals **160** and input signals **162** can be an analog signal or a digital signal. For example, an analog signal can be a voltage that varies between a low voltage to a high voltage whereas digital signals can be a series of discrete voltage states distributed over a voltage range. Operatively, engine control signals **160** cause various components of gas turbine engine **10** to change state or position. For example, engine control signals **160** can be used to vary the position of one or more fuel valves to vary the fuel flow rate entering a combustor. Other examples of engine control signals **160** include signals associated with engagement or disengagement of clutches and an angular position of variable vane stages. Input signals **162** are representative of one of engine parameters, aircraft parameters, and environmental parameters. Exemplary engine parameters include rotational speed of a low pressure spool, high pressure spool, boost spool, and/or fan shaft, the state or position of fuel valves, bleed valves, the state or position of clutch assemblies, the temperature or pressure within the compressor, combustor, or turbine, and engine power. Aircraft parameters include various parameters associated with an aircraft such as power lever angle, altitude, pitch angle, yaw angle, roll angle, rate of climb, and airspeed, among other possible parameters. Exterior parameters include ambient temperature and pressure at the inlet of gas turbine engine **10**.

(38) FIG. **7A** and FIG. **7B** are schematic views depicting an exemplary embodiment of gas turbine

engine **10** that includes transmission **400** rotationally coupling boost spool **12** and accessory gearing **402** to low pressure spool **14**. FIG. 7B is a section view of gas turbine engine taken along line C-C in FIG. 7A. Transmission **400** includes shaft gear **404**, inner tower gear **406**, tower shaft **408**, clutch **410**, shaft **412**, outer tower gear **412**, boost gear **414**, shaft **416**, and clutch **418**.

(39) Shaft gear **404** mounts to shaft **22** of low pressure spool **14**. Inner tower gear **406** mounts to tower shaft **408**, which extends radially outward with respect to shaft **22** from inner tower gear **406** to clutch **410**. Shaft gear **404** meshes with inner tower gear **406** to rotationally couple shaft **22** of low pressure spool **14** to tower shaft **408**. Shaft **412** extends from clutch **410** to output gear **413**. Boost gear **414** meshes with output gear **413** to rotationally couple shaft **412** to shaft **416**. Shaft **416** extends from boost gear **414** to clutch **418**. Boost shaft **76** extends along boost axis **88** to clutch **418**.

(40) Clutch **410** translates between an engaged state and a disengaged state to selectively couple tower shaft **408** to shaft **412**. In an engaged state, clutch **410** rotationally couples tower shaft **408** to shaft **412** such that shaft **22** of low pressure spool **14** is rotationally coupled to output gear **413**. In a disengaged state, clutch **410** rotationally uncouples tower shaft **408** from shaft **412** allowing shaft **22** of low pressure spool **14** to rotate independently of outer tower gear **412**.

(41) Clutch **418** translates between an engaged state and a disengaged state to selectively couple boost shaft **76** from boost gear **414**. In the engaged state, clutch **418** rotationally couples boost shaft **76** to boost gear **414** such that boost gear **414** and shaft **76** rotate together. In the disengaged state, clutch **418** uncouples boost shaft **76** and boost gear **414** such that boost spool **12** can rotate independently from boost gear **414**.

(42) Accessory gearing **402** is rotationally coupled to outer tower gear **412** and/or boost gear **414**. Accessory gearing **402** includes one or more gears enclosed within housing **420** of accessory gearbox **422** along with outer tower gear **412** and boost gear **414**. One or more gears of accessory gearing **402** are driven by outer tower gear **412** or boost gear **414**. At least some of the gears of accessory gearing **402** are rotationally coupled to respective output shafts **424**. Each output shaft **424** drives an engine accessory (e.g., a fuel pump, generators, constant speed drives, lubricating oil pumps, hydraulic pumps, and engine starters).

(43) Clutch **410** and clutch **418** can assume an engaged state or a disengaged state independently from the other. With clutch **410** engaged and clutch **418** disengaged, low pressure spool **14** drives accessory gearing **402** and, as such, provides power for engine accessories. Disengaging clutch **410** and engaging clutch **418** drives accessory gearing **402** and engine accessories with boost spool **12**. When both clutches **410** and **418** are engaged, low pressure spool **14** and boost spool **12** drive accessory gearing **402** and engine accessories, and when both clutches **410** and **418** are disengaged, no power is provided to accessory gearing **402** and engine accessories.

(44) Selective operation of boost spool **12** in concert with gas turbine engine **10** and transmission **400** depends on the operational range of gas turbine engine **10** input by power lever angle (PLA). The power lever angle (PLA) varies between zero percent (i.e., PLA **0**) and one hundred percent (i.e., PLA **100**). A power lever angle (PLA) equal to zero corresponds with an engine shutdown condition whereby fuel flow to combustor **70** stops and any remainder fuel is burned off or returned to fuel tanks of the aircraft. In order of increasing power level, intermediate power lever angles include engine start, ground idle and/or flight idle, a reduced cruising power level associated with loiter, cruise, and climb. A power lever angle (PLA) equal to one hundred percent corresponds to takeoff power. While power lever angle (PLA) values can be engine and/or aircraft dependent, exemplary power lever angle (PLA) values can be shutdown (PLA=0), engine start (0<PLA<5), ground idle and/or flight idle (5≤PLA≤10), loiter (30≤PLA≤40), cruise (40≤PLA≤60), climb (70≤PLA≤90), and takeoff (PLA=100).

(45) Based on power lever angle (PLA) and one or more additional engine parameters, aircraft parameters, and/or external parameters, controller **150** transmits control signals to components of transmission **400** as well as fuel system components regulating fuel flows to combustor **70** (i.e., a

primary combustor) and combustor **84** (i.e., a secondary combustor) to operate gas turbine engine **10** as described in the following scenarios.

(46) Within the operational power range, gas turbine engine **10** includes a minimum continuous power level and a maximum continuous power level. The minimum continuous power level and the maximum continuous power level correspond to the minimum and maximum power levels at which gas turbine engine **10** can operate continuously and in which pressure and temperature at various locations within the engine stabilize. For example, the minimum continuous power level can correspond to the ground idle condition while the maximum continuous power level can correspond to the takeoff conditions.

(47) Boost spool **12** operates within an intermediate power range of gas turbine engine **10**. The intermediate power range is a subset of sequential power levels of gas turbine engine **10** bound by a minimum power level and a maximum power level. The minimum power level of intermediate power range may coincide with power lever angles (PLA) corresponding to engine start, ground idle, flight idle, loiter, cruise, or a power level between engine start and cruise. The maximum power level of intermediate power range may coincide with power lever angles (PLA) corresponding to cruise, climb, takeoff, or a power level condition between cruise and takeoff power.

(48) In some embodiments, the intermediate power range can be further divided into two or more power subranges. For instance, sequentially continuous low power subrange and high power subrange can define the intermediate power range. The low power subrange extends from the minimum power level of intermediate power range to an intermediate power level greater than the minimum power level and less than the maximum power level of intermediate power range. The high power subrange extends from the intermediate power level to the maximum power level of the intermediate power range.

(49) Table 1 describes the operational range of gas turbine engine **10** equipped with transmission **400**. In this configuration, the intermediate power range extends to include the cruise condition and includes at least two power subranges. For example, the intermediate power range can extend from ground idle power corresponding to approximately ten percent power level angle (PLA=10) to cruise power corresponding to approximately sixty percent power level angle (PLA=60). Low power subrange extends from ground idle power to loiter power corresponding to approximately forty percent power lever angle (PLA=40). High power subrange extends from loiter power to cruise power.

(50) TABLE-US-00001

TABLE 1	Operational Profile-Multiple Clutch Transmission	PLA	Fuel Flow
Fuel Flow (% of Boost Tower Primary Secondary Condition Max)	Clutch State	Clutch State	
Combustor Combustor LP VAT Start/Ground 0-5 Engaged Disengaged f(EPRmin) f(NAGB) Open			
Idle Taxi/Low 10-30 Engaged Disengaged f(PLA) f(NAGB) Nominal Power Loiter/Cruise 40-60			
Engaged Engaged f(EPRmin) f(PLA) Open Takeoff/Climb 70-100 Disengaged Engaged f(PLA)			
None Nominal			

(51) At the engine start condition and up to ground idle power, controller **150** varies the fuel flow rate delivered to primary combustor **70** based on maintaining a minimum engine pressure ratio (EPRmin), or the ratio of exhaust pressure to inlet pressure. At the same time, controller **150** varies the fuel flow rate delivered to secondary combustor **84** based on a rotational speed of accessory gearbox **422**. Clutch **418** is engaged, rotationally coupling boost spool **12** to accessory gearbox **422**, and clutch **410** is disengaged, rotationally uncoupling low pressure spool **14** from accessory gearbox **422**. Variable area turbine **38** are commanded to or towards an open position, increasing the open area through variable area turbine **38** relative to the nominal or closed positions.

(52) As power level increases past ground idle power and up to loiter power, controller **150** supplies fuel to primary combustor **70** at a fuel flow rate varying as a function of the power lever angle (PLA). Secondary combustor **84** receives fuel at a rate that varies based on the rotational speed of the accessory gearbox **422** and in response controller **150**. Boost spool **12** continues to

drive accessory gearbox **422** via clutch **418**, which remains engaged. Clutch **410** remains disengaged and, consequently, low pressure spool **14** is rotationally uncoupled from accessory gearbox **422**. Controller **150** moves variable turbine vanes to or towards a nominal position, decreasing the open area through variable area turbine **38** relative to the open position.

(53) Between loiter power and cruise power, clutch **410** engages to rotationally couple low pressure spool to accessory gearbox. Within this power level range, controller **150** varies the fuel flow rate to primary combustor **70** based on a minimum engine pressure ratio (EPR_{min}) and varies the fuel flow rate to secondary combustor **84** based on the position of the power lever (PLA). Accordingly, the combined power delivered from low pressure spool **14** and boost spool **12** drive engine accessories attached to accessory gearbox. Controller **150** moves variable turbine vanes to or towards the open position increasing the open area through variable area turbine **38** relative to the nominal or closed position.

(54) At power levels greater than cruise power and up to takeoff power, controller **150** varies a fuel flow rate to primary combustor **70** based on the position of the power lever (PLA). While operating between cruise power and takeoff power, controller **150** stops fuel flow to secondary combustor **84** and disengages clutch **418**. As such, boost spool **12** is rotationally uncoupled from accessory gearbox and low pressure spool **14**. In response to control signals from controller **150**, variable turbine vanes move to or towards the nominal position.

(55) FIG. **8** depicts an exemplary speed profile of gas turbine engine **10** equipped with transmission **400**. As with the prior exemplary speed profiles, rotational speeds of high pressure spool **16** (NHP), low pressure spool **14** (NLP), boost spool **12** (NB), and the input of accessory gearbox **422**

(NAGB) are expressed as a percentage of respective maximum operational speeds of each spool or component. Dashed lines indicate starting, shutdown, or unpowered operation such as when a spool or component freewheels. Solid lines indicate operation of gas turbine engine **10** while delivering fuel to primary combustor **70**, secondary combustor **84**, or both primary and secondary combustors.

(56) As shown, the rotational speed of accessory gearbox **422** (NAGB) remains constant within the low power subrange in proportion to the rotational speed of boost spool. This can be achieved by varying fuel flow deliver to secondary combustor **84** of boost spool **12** to maintain a constant speed, or to minimize speed variation between ground idle power and loiter power. Low pressure spool **14** (NLP) accelerates independently of boost spool between ground idle power and loiter power by disengaging clutch **410**. At power levels greater than loiter power and up to cruise power, low pressure spool, boost spool, and accessory gearbox are coupled, respective speeds of each spool or component increasing in fixed relation as power lever angle increases. Between cruise power and takeoff power, low pressure spool operates without boost spool input. The gear ratio between boost spool **12** and low pressure spool **14** can be selected such that boost spool **12** attains its maximum operating speed when low pressure spool **14** attains approximately eighty percent to ninety percent of its maximum operating speed. Furthermore, the maximum operating speed of boost spool **12** coincides with the cruise condition and thereby provides increases the maximum overall pressure ratio (OPR) of gas turbine engine **10** during the longest phases of flight.

Discussion of Possible Embodiments

(57) The following are non-exclusive descriptions of possible embodiments of the present invention.

(58) A gas turbine engine according to an exemplary embodiment of this disclosure includes, among other possible things, a first spool, a second spool, a third spool, an accessory gearbox, a plurality of engine accessories, and a tower shaft. The first spool includes a first compressor and a first turbine mounted to a first shaft. The second spool includes a second compressor and a second turbine mounted to a second shaft. The third spool includes a third compressor and a third turbine mounted to a third shaft. The accessory gearbox includes a housing, a transmission gearing, a plurality of output shafts, and an accessory gearing rotationally coupled to the plurality of output shafts. The transmission gearing and the accessory gearing are enclosed with the housing. Each

engine accessory of the plurality of engine accessories is rotationally coupled to one or the plurality of output shafts. The tower shaft includes an inner gear enmeshing with the shaft gear and extends radially outward from the third shaft to the transmission gearing. An outer gear of the tower shaft enmeshes with the transmission gearing. The transmission gearing enmeshes with the accessory gearing and rotationally couples the tower shaft to the second shaft.

(59) The gas turbine of the preceding paragraph can optionally include, additionally and/or alternatively, any one or more of the following features, configurations, and/or additional components.

(60) A further embodiment of the foregoing gas turbine engine can include a first clutch between the tower shaft and the transmission gearing.

(61) A further embodiment of any of the foregoing gas turbine engines, wherein the first clutch can include a disengaged state whereby the transmission gearing is uncoupled from the tower shaft.

(62) A further embodiment of any of the foregoing gas turbine engines, wherein the first clutch can include an engaged state whereby the transmission gearing rotationally couples to the tower shaft.

(63) A further embodiment of any of the foregoing gas turbine engines can include a second clutch between the transmission gearing and the second shaft.

(64) A further embodiment of any of the foregoing gas turbine engines, wherein the second clutch can include a disengaged state whereby the transmission gearing is uncoupled from the second shaft.

(65) A further embodiment of any of the foregoing gas turbine engines, wherein the second clutch can include an engaged state whereby the transmission gearing is coupled to the second shaft.

(66) A further embodiment of any of the foregoing gas turbine engines can include a variable area turbine disposed between the first turbine and the third turbine.

(67) A further embodiment of any of the foregoing gas turbine engines, wherein vanes of the variable area turbine are rotatable about respective pivot axes to vary an open area of the variable area turbine.

(68) A further embodiment of any of the foregoing gas turbine engines, wherein a second maximum speed is the maximum continuous rotational speed of the second spool.

(69) A further embodiment of any of the foregoing gas turbine engines, wherein a third maximum speed is the maximum continuous rotational speed of the third spool.

(70) A further embodiment of any of the foregoing gas turbine engines, wherein a gear ratio between the second spool and the third spool is constant and rotates the second spool at the second maximum speed when a third rotational speed of the third spool is equal to or less than 90% of the third maximum speed and greater than or equal to 80% of the third maximum speed.

(71) A further embodiment of any of the foregoing gas turbine engines, wherein the housing of the accessory gearbox can subtend the gas turbine engine about the first spool and the third spool.

(72) A further embodiment of any of the foregoing gas turbine engines, wherein the third shaft can be concentric with the first shaft.

(73) A further embodiment of any of the foregoing gas turbine engines, wherein the second shaft can be offset from the first shaft and the third shaft.

(74) A further embodiment of any of the foregoing gas turbine engines can include a primary combustor communicating with the first compressor and the first turbine.

(75) A further embodiment of any of the foregoing gas turbine engines can include a secondary combustor communicating with the second compressor and the second turbine.

(76) A further embodiment of any of the foregoing gas turbine engines, wherein a primary flow path can include, in flow series, the third compressor, the first compressor, the primary combustor, the first turbine, and the third turbine.

(77) A further embodiment of any of the foregoing gas turbine engines, wherein a secondary flow path includes, in flow series, the second compressor, the secondary combustor, and the second turbine.

- (78) A further embodiment of any of the foregoing gas turbine engines, wherein the secondary flow path can intersect the primary flow path between the first compressor and the first turbine.
- (79) A further embodiment of any of the foregoing gas turbine engines, wherein the first compressor can be a high pressure compressor and the second compressor can be a low pressure compressor.
- (80) A further embodiment of any of the foregoing gas turbine engines, wherein the first turbine can be a high pressure turbine and the second turbine can be a low pressure turbine.
- (81) A method of operating a gas turbine engine includes, among other possible things, supplying a first fuel flow rate to a primary combustor of the gas turbine engine contemporaneously with supplying a second fuel flow rate to a secondary combustor of the gas turbine engine based on a power lever angle indicative of a first power subrange of an intermediate power range of the gas turbine engine. The intermediate power range consists of the first power subrange and a second power subrange that is less than a maximum continuous power and greater than a minimum continuous power associated with an operating range of the gas turbine engine.
- (82) The method of the preceding paragraph can optionally include, additionally and/or alternatively, any one or more of the following features, configurations, additional components, and/or steps.
- (83) A further embodiment of the foregoing method can include disengaging a first clutch uncoupling a tower shaft from an accessory gearbox based on the power lever angle less than the second power subrange.
- (84) A further embodiment of any of the foregoing gas turbine engines can include engaging a second clutch coupling a second shaft of a boost spool to an accessory gearbox based on the power lever angle greater than the minimum continuous power up to and including the intermediate power range.
- (85) A further embodiment of any of the foregoing gas turbine engines can include supplying a third fuel flow rate of the primary combustor of the gas turbine engine contemporaneously with supplying a fourth fuel flow rate to the secondary combustor of the gas turbine engine based on the power lever angle indicative of the second power subrange of the intermediate power range.
- (86) A further embodiment of any of the foregoing gas turbine engines, wherein the third fuel flow rate can vary to maintain a minimum exhaust pressure ratio of the gas turbine engine.
- (87) A further embodiment of any of the foregoing gas turbine engines, wherein the fourth fuel flow rate can vary based on the power lever angle indicative of the second power subrange of the intermediate power range.
- (88) A further embodiment of any of the foregoing gas turbine engines can include engaging the first clutch coupling the tower shaft to the accessory gearbox based on the power lever angle greater than the first power subrange.
- (89) A further embodiment of any of the foregoing gas turbine engines can include supplying the first fuel flow rate to the primary combustor of the gas turbine engine contemporaneously with stopping fuel flow to the secondary combustor based on the power lever angle greater than the second subrange and less than or equal to the maximum continuous power.
- (90) A further embodiment of any of the foregoing gas turbine engines, wherein the first fuel flow rate can vary based on the power lever angle.
- (91) A further embodiment of any of the foregoing gas turbine engines can include disengaging the second clutch uncoupling the second shaft of the boost spool from the accessory gearbox based on the power lever angle greater than the second power subrange and less than or equal to the maximum continuous power.
- (92) A further embodiment of any of the foregoing gas turbine engines can include supplying the second fuel flow rate to the secondary combustor contemporaneously with supplying the third fuel flow rate to the primary combustor based on the power lever angle less than the first power subrange and greater than or equal to the minimum continuous power.

- (93) A further embodiment of any of the foregoing gas turbine engines, wherein the second fuel flow rate can vary based on a rotational speed of an accessory gearbox.
- (94) A further embodiment of any of the foregoing gas turbine engines, wherein the third fuel flow rate can vary to maintain the minimum exhaust pressure ratio of the gas turbine engine.
- (95) A further embodiment of any of the foregoing gas turbine engines can include increasing an open area of a variable area turbine based on the power lever angle indicative of the second power subrange of the intermediate power range.
- (96) A further embodiment of any of the foregoing gas turbine engines can include decreasing an open area of the variable area turbine based on the power lever angle greater than the second power subrange and less than or equal to the maximum continuous power.
- (97) A further embodiment of any of the foregoing gas turbine engines can include increasing an open area of a variable area turbine based on the power lever angle less than the first power subrange and greater than or equal to the minimum continuous power.
- (98) A further embodiment of any of the foregoing gas turbine engines can include decreasing an open area of a variable area turbine based on the power lever angle indicative of the first power subrange.
- (99) A further embodiment of any of the foregoing gas turbine engine, wherein the variable area turbine can be disposed between a first turbine and a third turbine of the gas turbine engine.
- (100) A further embodiment of any of the foregoing gas turbine engines can include adjusting the open area of the variable area turbine based on the rotational speed of the first turbine.
- (101) While the invention has been described with reference to an exemplary embodiment(s), it will be understood by those skilled in the art that various changes may be made and equivalents may be substituted for elements thereof without departing from the scope of the invention. In addition, many modifications may be made to adapt a particular situation or material to the teachings of the invention without departing from the essential scope thereof. Therefore, it is intended that the invention is not limited to the particular embodiment(s) disclosed, but that the invention will include all embodiments falling within the scope of the appended claims.

Claims

1. A method of operating a gas turbine engine having an operational power range that includes an intermediate power range consisting of a first power subrange and a second power subrange that is less than a maximum continuous power and greater than a minimum continuous power, the method comprising: supplying a first fuel flow rate to a primary combustor of the gas turbine engine contemporaneously with supplying a second fuel flow rate to a secondary combustor of the gas turbine engine based on a power lever angle indicative of the first power subrange of the intermediate power range, wherein: the first fuel flow rate varies based on the power lever angle; and the second fuel flow rate varies based on a rotational speed of an accessory gearbox, wherein the second fuel flow rate varies to minimize variation of the rotational speed of the accessory gearbox within the first power subrange, towards substantially achieving a constant rotational speed of the accessory gearbox; disengaging a first clutch uncoupling a tower shaft from the accessory gearbox based on the power lever angle less than the second power subrange; and engaging a second clutch coupling a second shaft of a boost spool to the accessory gearbox based on the power lever angle greater than the minimum continuous power up to and including the intermediate power range; wherein disengaging the first clutch and engaging the second clutch occur before supplying the first fuel flow rate to the primary combustor and supplying the second fuel flow rate to the secondary combustor.
2. The method of claim 1, further comprising: supplying a third fuel flow rate to the primary combustor of the gas turbine engine contemporaneously with supplying a fourth fuel flow rate to the secondary combustor of the gas turbine engine based on the power lever angle indicative of the

second power subrange of the intermediate power range, wherein: the third fuel flow rate varies to maintain a minimum exhaust pressure ratio of the gas turbine engine; and the fourth fuel flow rate varies based on the power lever angle indicative of the second power subrange of the intermediate power range; and engaging the first clutch coupling the tower shaft to the accessory gearbox based on the power lever angle greater than the first power subrange.

3. The method of claim 2, further comprising: supplying the first fuel flow rate to the primary combustor of the gas turbine engine contemporaneously with stopping fuel flow to the secondary combustor based on the power lever angle greater than the second subrange and less than or equal to the maximum continuous power, wherein the first fuel flow rate varies based on the power lever angle.

4. The method of claim 3, further comprising: disengaging the second clutch uncoupling the second shaft of the boost spool from the accessory gearbox based on the power lever angle greater than the second power subrange and less than or equal to the maximum continuous power.

5. The method of claim 4, further comprising: supplying the second fuel flow rate to the secondary combustor contemporaneously with supplying the third fuel flow rate to the primary combustor based on the power lever angle less than the first power subrange and greater than or equal to the minimum continuous power, wherein: the second fuel flow rate varies based on the rotational speed of the accessory gearbox; and the third fuel flow rate varies to maintain the minimum exhaust pressure ratio of the gas turbine engine.

6. The method of claim 5, further comprising: increasing an open area of a variable area turbine based on the power lever angle less than the first power subrange and greater than or equal to the minimum continuous power, wherein the variable area turbine is disposed between a first turbine and a third turbine of the gas turbine engine.

7. The method of claim 4, further comprising: decreasing an open area of a variable area turbine based on the power lever angle greater than the second power subrange and less than or equal to the maximum continuous power, wherein the variable area turbine is disposed between a first turbine and a third turbine of the gas turbine engine.

8. The method of claim 2, further comprising: increasing an open area of a variable area turbine based on the power lever angle indicative of the second power subrange of the intermediate power range, wherein the variable area turbine is disposed between a first turbine and a third turbine of the gas turbine engine.

9. The method of claim 8, further comprising: adjusting the open area of the variable area turbine based on a rotational speed of the first turbine.

10. The method of claim 1, further comprising: decreasing an open area of a variable area turbine based on the power lever angle indicative of the first power subrange, wherein the variable area turbine is disposed between a first turbine and a third turbine of the gas turbine engine.

11. The method of claim 1, wherein the second fuel flow rate is varied to maintain the rotational speed of the accessory gearbox constant within the first power subrange.
